

EDU430

ICT in Education

Virtual University of Pakistan, Lahore

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Topic No. 1

Definition of ICT

ICT is concerned with the storage, retrieval, manipulation, transmission or reception of digital data.

Communication Technology

- Communication Technology implies the knowledge, skills and understanding needed to exchange information verbally or nonverbally.
- It is processing of information in terms of accessing information, decoding information and sending it via a medium.
- Communication Technology is the electronic systems used for communication between individuals or groups.

Topic No. 2

Information and Communication Technology

"ICT is defined, as a diverse set of technological tools and resources used to communicate, and to create, disseminate, store, and manage information" (Mangal, 1997).

Characteristics of ICT

- ICT is the integration of Hardware and Software Technology.
- ICT fosters the dissemination of information and knowledge.
- The nature of ICT is digital and virtual.
- ICT is global, anywhere, any time.
- ICT is a key enabler in the creation of networks.

Topic No. 3

Computer Informatics

- Informatics" is the study of information processing
- Educational Informatics is the study of the application of digital technologies and techniques to the use and communication of information in learning and education.
- Educational Informatics is the development, use, and evaluation of digital systems that use pedagogical knowledge to engage in or facilitate resource discovery in order to support learning.
- Studying the techniques and impact of „digital technologies“ within the context of the „use and communication of information“ in educational settings.
- Educational informatics represents the intersection of three disciplines: teaching and learning, information science, and information communication technologies (ICTs).

Computer Network

- Inter connected collection of autonomous computers.

Topic No. 4

Computer Networking

- Copper wire
- Microwaves
- Fiber optics
- Communication satellites

Local Area Network and its Characteristics

- Local area networks are networks with in a single building or campus or up to a few kilometers in size.

Characteristics of LAN

- Low delay
- Very few errors
- Higher speed
- LANs are restricted in size (Functions within 10 KMs).
- Transmission time is bounded.
- Simple network management

Topic No. 5

Metropolitan Area Network (MAN)

- MAN is an extension of LANs.
- It normally covers the area of a city (are less than 100KMs).
- Uses a combination of different hardware and transmission media to cover distances efficiently.

Wide Area Network and its Characteristics

- It is defined as networking in a large geographical area, often a country or continent.

Characteristics of WAN

- WAN uses telephone lines or satellite links as a medium.
- LAN may be connected to WAN.
- It covers different cities or countries.
- WANs are not owned by anyone organization but are under collective ownership and management.
- Complex network design
- Error rate is large.
- Speed is less compared to LAN

Advantages of ICT in Education

- Quick access to information
- Easy availability of updated data
- Connecting Geographically dispersed regions
- Catering to the Individual differences
- Wider range of communication media
- Wider learning opportunities for pupils

Topic No. 6

Internet

Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words „interconnection“ and „network“. Through this, each computer user can communicate and share common resources with other users.

Web Server

It is a programme and a computer clubbed into one entity which responds to the requests from web browsers for internet resources. The primary function of a web server is to deliver web pages to clients“ e.g. Apache HTTP Server, Microsoft Internet Information Services (IIS) and Sun Java System Web Server.

Topic No. 7

Web Browser

It is a programme that allows a user to display and interact with a „hypertext“ documents e.g. Internet Explorer, Net Scape Navigator, Mozilla Firefox, and Google Chrome.

Difference between Website and Web Page

Website

Website is defined as a collection of related pages on particular information.

Web Page

It is a file of information stored on a web server.

Topic No. 8

Homepage

Home page is the first hypertext document to be shown when a user follows a link to the web server.

Search Engines

Search engines can be defined as special tool to search for information on the net.

- AltaVista
- Yahoo
- Google
- Bing
- AOL

Topic No. 9

World Wide Web (WWW)

- WWW is a vast collection of documents stored on internet computers.
- It was invented by Tim Berner Lee.
- It provides easy access to information.
- It is a system of media rich documents spread across the internet, linked with each other.
- WWW documents contain texts, pictures, sound, video, etc.

Intranet and its Characteristics

- Intranet can be defined as collection of private computer networks within an organization. Internal or private network of an organization based on internet technology (such as hypertext and protocols) and accessed over the internet.
- An intranet is meant for the exclusive use of the organization and its associates (customers, employees, members, suppliers, etc.) and is protected from unauthorized access with security systems such as firewalls.
- An intranet should be closed off to the general public and require a login for authorized users.

Topic No. 10

Difference between Uploads and Downloads

- Both upload and download refer to the process of transferring a file.

Uploading

- Upload means to transfer a file or files from your own computer to another computer.
- Uploading is the process of copying files from your computer to a remote location. For example, you might transfer a file from your home PC to the Yahoo! Computer that stores your Web Hosting files.

Downloading

- Download means to transfer a file from another computer to your own computer.
- Downloading is the process of copying files to your computer from a remote location. For example, you might download a file from your Web Hosting account to your home PC.

Hosting

- For a website to be viewed by other people it must be stored on a computer (server) that is connected to the internet.
- A company that provides this service is known as a host; and the service it provides is hosting.

Uses of Computer Networks

- For sharing resources
- For high reliability
- For saving money
- For using as communication medium
- For accessing remote information
- For person to person communication
- For interactive entertainment

Topic No. 11**Web Resources**

Web resources are defined as web pages and documents on the Internet that provide useful information.

Bookmarks

- Book marks can be defined as, a collection of direct links to predefined web pages which are stored in your web browser.
- Bookmarks can be created and managed by the user themselves. Most browsers come packaged with several existing bookmarks.

Topic No. 12**Hypertext**

- Hypertext is generally the text displayed on a computer display or other electronic device.
- It may have links to other text which the reader can immediately access.
- The hypertext pages are interconnected by hyperlinks, typically activated by a mouse click, key press or by touching the screen.
- Hypertext is sometimes used to describe tables, images and other with hyperlinks.
- Hypertext is the underlying concept defining the structure of the World Wide Web with pages often written in the HTML.
- It enables an easy to use and flexible connection and sharing of information over the Internet.

Markup Language

- Markup language is a modern system for annotating a document in a way that is distinguishable from the text.
- The idea and terminology evolved from the "marking up" of paper manuscripts.
- A widely used markup language is Hyper Text Markup Language (HTML), one of the document formats of the World Wide Web.

Topic No. 13**Learning Aids**

- Learning aids can be defined as any device/Aids which help self-learning/group learning.

Teaching Aids

- Teaching aids are the physical tools used to convey information in the classroom.
- Teaching Aids are any device which can be used to make the communication more effective.
- Teaching Aids are any device which can be used to make the learning experience more concrete, more realistic and more dynamic.
- 50 % of what we HEAR and SEE
- 70 % of what we SAY
- 90 % of what we SAY and DO

Topic No. 14**Psychology of Teaching Aids**

Research done by COBUN (1968) indicates that generally we LEARN:

- 1 % through TASTE
- 1.5 % through TOUCH
- 3.5 % through SMELL
- 11 % through HEARING
- 83 % through SIGHT

Research done by COBUN (1968) indicates that generally we remember:

- 10 % of what we READ
- 20 % of what we HEAR
- 30 % of what we SEE

Advantages of Teaching Aids

- Help to learn more and remember for long.
- Motivate the learner.
- Give clarity to learning.
- Give reality and vividness to learning situations.
- Make abstract concrete
- Reduce verbalism.
- Provide variety in methods of teaching and learning.
- Make learning interesting meaningful and permanent.
- Develop deeper understanding.
- Arouse curiosity and self-activity.
- Saving of energy and time
- Spread of education on a mass scale.

Topic No. 15

Characteristics of Good Teaching Aids

- They should be meaningful and purposeful
- They should be accurate in every respect.
- They should be simple.
- They should be cheap.
- As far as possible, they should be improvised.
- They should be large enough to be properly seen by the students for whom they are meant.
- They should be up to date.
- They should be easily portable.
- They should be according to the mental level of the student.
- They should be motivating the learner.

Programmed Learning

- Programmed learning is a self-instructional and self-corrective technique in which all the learning material is presented to the learner stage by stage through sequentially arranged smaller units called frames. The frames are graded according to the level of difficulty as well as logical sequence.
- Programmed Learning is a carefully specified, systematically planned, empirically established, skillfully arranged and effectively controlled self-instructional technique for providing individualized instruction or learning experience to the learner.
- Programmed learning is available in the form of books, cards, machine, linear programming, branched programming etc.

Hyper Text Markup Language (HTM)

- Hyper Text Markup Language (HTML) is a standard language used for developing web pages". Hyper Text Markup Language is a markup language that web browsers use to interpret and compose text, images and other material into visual web pages.

History of HTML

- In 1980, a markup language, i.e., Standard Generalized Markup Language or SGML was developed.
- HTML was evolved from SGML.
- Presently, HTML is advanced to DHTML and then to XML (Extensible Markup Language) and now to VRML (Virtual Reality Modeling Language).

HTML Coding

- HTML codes are known as „HTML Tags“.
- An HTML document has two parts, Markup Language Tags and Web page content Text Information.

Points to Remember While Writing HTML Document

- Each tag is enclosed within a „<“ (left bracket) and „>“ (right bracket). This differentiates them from text.
- An opened tag must be closed

- Opening tag is denoted by < >
- Closing tag is denoted by </ >
- Generally the HTML tags are written in upper case
- It is not a case sensitive language.
- Coding can be done in any text Editor (Windows notepad, Ubuntu –G-Edit).
- When we save HTML file, we should add .html or .HTM after the file name.

Topic No. 16

Teaching as a Noble Profession

- Teaching is regarded as the noblest of all professions in the world.
- A good teacher not only shows the right path that the students should follow but also prepares the human resource for the further development of the nation.
- Teaching has the power to create a healthy and peaceful world.
- It has the capacity to inspire lifelong learning and a passion for knowledge, understanding, and innovation.

Teaching

- “Teaching is an intimate contact between a more mature personality and less mature one which is designed to further the education of the latter.”
- The aim of teaching is to transform a total man and his personality

Training

- In training the various kinds of skill are taught.
- Training has specific goals of improving one's capacity, and performance.
- The trainees are taught to perform skills which they attain during training and practice.
- There is mental and physical activity in training, but more and more of the activity is physical.

Instruction

- Instruction is at the higher level of training.
- Instructions tell people how to perform a process.
- In instruction there is better level of the use of intellectual powers of the trainee.
- The aim of teaching is to transform a total man and his personality, but the aim of instruction is to develop the intellectual powers in a given direction.

Topic No. 17

Computer Assisted Instruction (CAI)

- A self-learning technique, usually offline/online, involving interaction of the student with programmed instructional materials.
- “It is the use of computer to assist in the presentation of instructional material to a student to monitor learning process or to select additional instructional material in accordance with the needs of individual learners”
- CAI facilitates instant access to accurate information with infinite opportunity for practice. It also provides opportunity for systematically organized learning, to the maximum level possible, for all learners. It makes learning easy and interesting.
- Computer Assisted Instruction (CAI) is an interactive instructional technique whereby a computer is used to present the instructional material.
- CAI uses a combination of text, graphics, sound and video in enhancing the learning process.

Teaching Machines

- These were developed by S.L. Pressy.
- It's a piece of device designed to be operated by an individual student for self-learning.
- The student is presented with a question by some form of display on the machine.
- The student is required to respond either to write answer or put a button to indicate the correct response.
- The student will inform the correctness of the answer.
- An account is kept of the responses made by the student.

Topic No. 18

Information and Communication Technology (ICT)

This concept involves transfer and use of all kinds of information. ICT is the foundation of economy and a driving force of social changes in the 21st century. Distance is no longer an issue when it comes to accessing information. For example, working-from home, distance learning, e-banking, and e-government are now possible from any place with an Internet connection and a computing device.

Information Technology

IT (Information Technology) encompasses all of the technology that we use to collect, process, protect and store information. It refers to hardware, software (computer programs), and computer networks.

Topic No. 19**Hardware**

The concept of hardware includes computer components, the physical and tangible parts of the computer, i.e., electrical, electronic and mechanical parts which comprise a computer.

Working Principle of a Computer

Computer working principle: data are entered into a computer via input devices, then are processed and stored in a system unit, and are finally displayed by the output device.

Topic No. 20**Personal Computer**

Personal computer (PC), as the name suggests, is intended for personal use, as opposed to the server, which is used by a larger number of people simultaneously, from different locations, often via terminals. If you do not intend to move your computer frequently from one place to another, and at the same time you want maximal price/performance ratio, then you should use a desktop computer. In comparison to laptops or tablet computers, it is much larger in size, inconvenient to carry/move, consumes more electricity but has a much better price/performance ratio. Also, they are much easier to upgrade.

Desktop Computers and Laptop or Tablet PC

- Laptop or tablet PC is used by individuals who have the need to travel with a computer or simply use them for aesthetic reasons when computing power is not an issue. Laptop computers, as opposed to tablet PCs, more closely resemble a personal computer when it comes to data input. Data entry is done via keyboard and mouse, while the tablet PC data entry is done via touch screen.
- Unlike desktop computers, notebooks and tablet PCs are optimized for portability, low power requirements at the expense of performance and can be used (for a limited period of time-i.e. until the batteries are depleted) without connection to the power grid. In order to prepare a laptop or a tablet computer for use without a power connection, it is necessary to recharge the batteries.

Meaning of Teaching

- Teaching is to cause the child to learn and acquire the desired knowledge, skills and also desired ways of living in the society.
- Teaching is the stimulation, guidance, direction and encouragement of learning.
- Teaching is the communication between two or more persons, who influence each other by their ideas and learn something in the process of interaction.
- Teaching is to fill the minds of the learner by information and knowledge of facts for future use.
- Teaching is the process in which learner, teacher and other variables are organized in a systematic way to attain some predetermined goals.

Definitions of Teaching

- "Teaching is an intimate contact between a more mature personality and less mature one which is designed to further the education of the latter." H.C.Morrison (1934)
- "Teaching means many different things, that teaching act varies from person to person and from situation to situation." (Bar, 1961)
- The behavior or activities of persons as they go about doing whatever is required of teachers, particularly those activities which are concerned with the guidance or direction of learning of others." (Ryan, 1965)
- Teaching is the arrangement of contingencies of reinforcement under which students learn. They learn without teaching in their natural environment, but teachers arrange special contingencies which expedite learning and hastening the appearance of behavior which would otherwise be acquired slowly or making scene of the appearance of behavior which might otherwise never occur." (B.F. Skinner. 1968)
- "Teaching is an act of interpersonal influence aimed at changing the ways in which other persons can or will behave." (N.L. Gage, 1963)

Functions of Teaching**Preservative Function of Teaching**

- The main function of TEACHING is to preserve and protect the old values, customs, beliefs, traditions, etc.
- Teaching must help in preserving the vital elements of our heritage.

Transmittive Function of Teaching

- Teaching transfers knowledge, ideologies, theories, principles and attributes from one generation to another. Thus, it contributes greatly to the general progress of society.

Progressive Function of Teaching

- Teaching provides opportunities to recognize and restructure human experiences and as a result of this, there is development of human civilization and culture.

Topic No. 21**PDA (Personal Digital Assistant) PALM**

- PDA Personal Digital Assistant (PALM) is a convenient small sized computer. It easily connects to mobile phones and can prove a good solution for less demanding users. As the name suggests, it is a device that fits in the user's palm. Its name directly tells us that this computer is more of an assistant and not a workstation-whose name suggests the superiority in capabilities and computing power, especially in comparison with PDA.

Mobile Phones and Smart Phones

- Mobile phone is a portable electronic device used for distant communication. In recent years, mobile phone has evolved from simple communication device into a multifunctional device. Additional functions, such as short text messaging (SMS), electronic mail, Internet access, contact registration, calculator, clock, alarm, recording and photograph displaying, recording and playback of video clips, sending/receiving multimedia messages (MMS), audio recording and playback, etc. has turned the mobile phone into an extremely useful device, whose absence would make active involvement and participation in a modern society not possible.
- Smartphone is a device that merges functionality of phones, PDAs, cameras, camcorders and computers. To function properly, Smart phones use operating systems, which are the basis for application development. Some smart phones can be connected to an external screen and keypad, which creates a working environment, similar to that of a laptop or a desktop computer. Some operating systems for Smartphone are: Google Android, Symbian, Blackberry, Palm Pilot, and Windows Phone.

Topic No. 22**System Unit**

The system unit (case) contains a computer's vital parts. There are two basic types of cases:

- Desktop casing is placed on a desk in a horizontal orientation.
- Towers come in 3 sizes (mini-tower, mid-tower and full-tower) and it is vertically orientated.

Mother Board and Protocol

- Motherboard, MBO is computer's basic circuit, to which all computer components are connected, directly or indirectly. Devices are connected to the motherboard through a system bus. System bus connects all devices, ensures data flow and communication between different devices using predefined protocols.
- Protocol describes a manner in which communication between devices is defined. It enables them to address each other and defines how they should look for each other on either system bus or network. Buses can, according to the purpose, be divided into:
 1. Serial USB, Firmware, etc.
 2. Parallel AGP, PCI, etc.
 3. Mixed-Hyper Transport, Infini Band, PCI, etc.

Topic No. 23**Cache**

- Cache is a small capacity memory which allows quick access to data. By storing data from working memory in cache, the speed of communication between processor and RAM is increased. Microprocessors use three levels of fast cache, L1, L2 and L3, used to store often used data.

Central Processing Unit

- Central Processing Unit (CPU or processor) is a central part of a computer (and can be referred to as the computer's "brain"). It manages all other computer parts, monitors their mutual communication and performs arithmetic-logical operations. Processor speed is measured in hertz (or megahertz or gigahertz). Most famous manufacturers for personal computer processors are Intel and AMD.

Topic No. 24

Hard Disk Drive

- Hard Disk Drive (HDD) is a place for permanent data storage (it does not delete/clear when computer shuts down). Its features are: large capacity, faster performance in comparison to optical devices but slower in comparison to RAM and are used for permanent data storage. We can distinguish between internal and external hard drives.

Read Only Memory (ROM) and Random Access Memory (RAM)

- ROM (Read Only Memory) is a type of permanent, internal memory that is used solely for reading. All BIOS programs are stored in ROM.
- RAM (Random Access Memory) is a working memory in which analyzed data and programs are stored, while computer runs. It allows reading and writing data, and it is deleted/cleared when the computer shuts down.

Topic No. 25

BIOS

BIOS (Basic Input/output System), can be defined as a program which is located in a separate ROM on the motherboard, and defines, as the name suggests, basic input/output system, is a good example.

Floppy Disk Drive

Floppy Disk Drive is used for storing and reading data stored on a floppy disk. Disk capacity is 1.44MB. Before memory stick and a wider usage of CD recorders, it was used as data carrier. Modern memory sticks have a memory capacity measured in GB while floppy disks only have memory capacity of 1.44MB, indicating that floppy disks are becoming obsolete.

General Principles of Teaching Aids

Principle of Selection

- They should suit the age level, grade level and other characteristics of the learners.
- They should have educational value besides being interesting and motivating.
- They should be the true representation of the real things.
- They should help in the realization of desired learning objectives.

Principle of Preparation

- As far as possible, locally available materials should be used in the preparation of an aid.
- The teachers should receive some training in the preparation of aids.
- The teachers themselves should prepare some of the aids.
- Students may be associated in the preparation of aids.

Principle of Physical Control

- This principle relates to the arrangement of keeping aids safely and also to facilitate their lending to the teachers for use.

Principle of proper Presentation

- Teachers should carefully visualize the use of teaching aids before actual presentation.
- Adequate care should be taken to handle an aid in such a way as no damage is done to it.
- The aid should be displayed properly so that all the students are able to see it.
- As far as possible, distraction of all kind should be eliminated so that full attention may be paid to the aid.

Topic No. 26

Sound Card

Soundcard is a device used for sound creation and production by means of computer speakers.

CD-ROM and DVD Drive

- CD-ROM drive is used for reading CD media.
- DVD drive is used for reading DVD discs. DVD disc capacity ranges from 4.7 to 18GB.

Topic No. 27

Graphic Cards

Graphics card is responsible for image processing and displaying it on a monitor. It has its own graphics processor and memory. Image quality depends on the strength of these components.

Modem and Connectors

- Modem enables computers to communicate via telephone lines. They connect computers to the Internet.
- Connectors or ports are slots visible in the back and the front side of a computer.

Topic No. 28**Universal Serial Bus (USB)**

Universal Serial Bus (USB) is used to connect various devices (mouse, keyboard, USB memory).

Serial Port and Parallel Port

- Serial port is used for example in connecting a mouse (labeled COM1 or COM2).
- Parallel port is used for connecting a local printer (LPT1 or LPT2).

Topic No. 29**Network Port and Firmware**

- Network port is used for connecting computers to a network.
- Firmware is used for connecting computers and audio-video devices (digital cameras, etc.).

Measurement Units

Bit (binary digit) is the basic unit used to measure the amount of information. A byte or octet contains eight bits.

- 1 KB (kilobyte)- 1024 B (approx. 1000 B)
- 1 MB (megabyte)- 1024 KB (approx. 1000 KB)
- 1 GB (gigabyte) -- 1024 MB (approx. 1000 MB)
- 1 TB (terabyte) – 1024 GB (approx. 1000 GB)

Topic No. 30**Blu-ray Disc**

Blu-ray disc (BD)- the successor to DVD, is an optical disk storage, it comes in different capacities, depending on how many layers it has and the capacity of each layer. Currently, the capacity of one layer is between 27 GB and 33 GB, while the overall capacity is the product of the number of layers and capacity of each layer.

Memory Card and USB Stick

Memory card is a type of flash memory used to store data in digital cameras, cell phones, MP3 players etc. USB Stick is a data storage device. It features small dimensions, relatively high capacity, reliability and speed. It belongs to the type of flash memory that remembers data, even when not under voltage i.e. they do not need electric power to maintain data integrity.

Limitations of ICT

ICT supposedly endangers the child's social development.

- ICT is supposedly to endanger the child's intellectual development.
- Decreasing of child's emotional development
- Child's health is endangered.
- Misuse of ICT A
- Complete control over the child's usage of ICT is today
- Practically impossible
- Expensive

Topic No. 31**Internal and External Hard Drive**

- There is a difference between an internal hard disk drive, which is embedded in the computer case, and an external hard disk drive, which is connected to a computer by using an appropriate cable or USB port, and is usually used to transfer data from one computer to another or for backup.

Mouse

- Mouse is an input device that facilitates work with the graphical user interface (GUI). The mouse transmits hand movements and the screen displays the cursor (mouse pointer) movements. They are divided into mechanical and optical (with respect to a transfer movement), and wired and wireless (with respect to connection).

Topic No. 32

Track Ball and Keyboard

- Trackball, unlike a mouse, is not movable. Hand movements are transmitted to the screen by rolling the ball which is located on the upper side of the device.
- Keyboard is used for data entry and issuing commands. They can also be wired or wireless.

Scanner

- Scanner is used to load data (image, text, etc.) from the printed material into a computer. The result of scanning is an image, but with special programs, if we scan the text, we can get a text as a result. Software used to recognize text from image is called a text recognition tool.

Topic No. 33

Touch Pad and Light Pen

- Touchpad is used for transmission of hand movement, but unlike working with a mouse, the user is the one who determines the position of the cursor by touching the touchpad.
- Light pen enables handwriting on screen and can be used as a mouse. It requires an appropriate monitor type.

Joystick

Joystick: mainly used in computer games. Unlike a mouse, it has many buttons which allow control over game objects.

Topic No. 34

Microphone

Microphone is a device that converts sound into an electrical signal, which can be stored on a computer. It is mainly used for recording sound, communication between players in online games, in combination with a web camera in video conferencing, for converting voice into text on a computer (speech-to-text processing (e.g., textual files or emails), etc.

Structure of Monitor

Monitor displays images from the computer, it enables us to see, work and control computers. In other words, working on a computer without a monitor would be inconceivable. Common types of monitors, with regard to manufacturing technology, are the CRT and LCD. CRT monitors have been present on the market for a long time, and other technologies are pushing them out. They are based on cathode tube technology. LCD monitors use liquid crystal technology. In comparison with CRT monitors, LCD monitors use less electrical energy, do not emit radiation and their price is higher, however due to smaller dimensions, more attractive design and a good picture quality, they are pushing CRT monitors out of the market. Monitor size is expressed by the size of screen diagonal and measured in inches ("). Picture quality is expressed with the notion of resolution, which is a number of horizontal and vertical dots (pixels) (e.g. 1920x1080).

Topic No. 35

Printer

Printer is a device used for printing data from a computer onto a paper. We distinguish between local printer (connected directly to the computer) and network printer (connected directly to network using a network card). Also, printers also differ according to print technology: dot matrix, laser, inkjet, thermal printer and plotter.

Dot Printer and Laser Printer

Dot matrix printers are the oldest, with the lowest price of print per paper, they are slow, make a lot of noise while printing, and are mostly appropriate for printing text.

Laser printers are similar to photocopy devices when it comes to technology. They have exceptional print quality, speed and are quiet. Downsides of laser printers are their high price and high price of toners.

Merits of HTML

- HTML is simple but powerful formatting language to use.
- The simplicity allows anyone to create web pages.
- HTML documents are device independent (works on any platform)
- The web pages can be linked together using links
- The pages can be updated at any time.

Demerits of HTML

- HTML does not have any programming capabilities.
- Require expertise knowledge.
- Time consuming.
- Difficult to remember coding

Topic No. 36**Projector**

Projector is a device used to project a computer image or other images from independent devices, such as DVD players, Blu-ray player, etc. onto canvas or a wall.

Inkjet Printers and Thermal Printer

- Inkjet printers have a high print quality (somewhat lower in comparison with laser printer), they are quiet while printing, and have low initial investment. Ink price, especially color ink, can cost as much as the printer itself. Printing technology is based on ink dispersion from container onto paper.
- Thermal printer, as its name states, leaves a print on the paper by utilizing heat. They use paper sensitive to heat, feature small dimensions; they are quiet while printing and relatively cheap. They are usually used for printing receipts, and owing to that they are called POS printer (printer of sale). Also, they are used as calculator printers and due to their small dimensions, as portable printers.

Topic No. 37**Plotter**

Plotter is used for printing large drawings (up to A0). They are extremely expensive and used only for professional purposes, such as in designing firms for printing technical drawings (blueprints).

Storage Devices and Touch Screen

- Storage devices, due to necessity for writing and reading data, they are classified as input/output devices.
- Touch screen (i.e. monitor sensitive to touch) is out device while displaying computer image, and at the same time input device while receiving manual orders.

Topic No. 38**Software**

Software is, unlike hardware, intangible part of the computer. It consists of a sequence of commands, written according to strict rules. Programs are written by programmers, in various programming languages.

Operating System

Operating system is a program which manages computer hardware. First computers did not have operating systems; they had programs that were directly loaded into the computer (e.g. punch cards). Today, computers have an operating system which loads into the computer's memory during its startup. Computer functions are based on its operating system. Within operating system, drivers (responsible for the functioning of a computer) and various utility programs (responsible for the functionality of a computer) are installed. The most famous operating systems are:

- Linux (Debian, Ubuntu, Fedora, Knoppix) open source software
- Microsoft Windows (XP, Vista, 7) proprietary software
- Mac OS X (Cheetah, Panther, Snow Leopard) proprietary software

Topic No. 39**Application Software**

Application Software (Utility programs) are all programs that users use to perform different tasks or for problem solving. Users, according to his/her needs, install the appropriate utility software. Computer functions and tasks that computers can perform are defined by the installed utility software. Utility software can often cost more than computer hardware unless the software is open source.

Text Processing Software and Spread Sheet Calculation Software

- Text processing software is used for creating and forming text documents and nowadays, they can contain images, charts and tables. Examples of such programs are OpenOffice.org Writer (open source software) and Microsoft Word (proprietary software).
- Spreadsheet calculations software is used for performing various calculations and presentation of results in charts. Examples of such programs are Open Office, Writer (open source software) and Microsoft Excel (proprietary software).

Topic No. 40

Software for Presentation

- Software for presentations is used to create professional presentations that consist of slides with graphical and textual elements. Such a presentation can afterwards be displayed as a "slide show" by using a projector. Examples of such programs are OpenOffice.org Impress (open source software) and Microsoft PowerPoint (proprietary software).

Magnifier and On Screen Keyboard

- Magnifier is used to enhance a part of the screen.
- In on Screen Keyboard text is entered using a mouse to click on the on-screen keyboard.

Importance of Audio Visual Aids

- They are best motivators
- Reduce verbalism
- Gives clear images
- Vicarious experience
- Provide variety in methods of teaching and learning
- Freedom to child
- Opportunities to handle and manipulate
- Contributes to increased retentively
- Based on maxims of teaching
- Helpful in attracting attention
- Helpful in fixing up new learning
- Saving energy and time
- Gives reality and vividness to learning situations
- Meeting individual differences
- Encouragement to healthy classroom interaction
- Spread of education on a mass scale

Topic No. 41

Narrator

Narrator is commonly used by users with visual impairment - it can read text displayed on monitor, it tells current cursor position, and describes certain events, like warning and error messages generated by OS.

Function of Windows Speed Recognition

Windows Speech Recognition enables speech recognition, i.e. recognizes spoken word, transfers it to text and enters it into a document; therefore it enables you to dictate a text to a computer, to browse the web using your voice etc.

Topic No. 42

Computer Network

Computer network is comprised of at least two, connected, by wire or wireless, computers that can exchange data i.e. communicate. There are many reasons for connecting computers into a network, and some of them are:

- Exchange of data between users that have network access,
- Access to shared devices, such as network printers, network disks, etc.,
- Enables user communication and socializing, etc.

LAN (Local Area Network)

It is defined as a network that covers a relatively small geographical area- it connects computers within a firm or household by wire.

WLAN (Wireless Local Area Network)

It is defined as, a network that covers a relatively small geographical area – it connects computers within a firm or household wirelessly.

WAN (Wide Area Network)

It is defined as, a network that covers a relatively large geographical area – it connects a greater number of computers and local networks.

Topic No. 43

Client and Server

Relationship client - server is defined in the following manner: client sends requests and server responds to those requests. We can use Internet as the best known example. User's computer, connected to the Internet, sends requests to a certain web page (by entering page address into the Internet browser Address bar), and the server responds. Web page is loaded into the user's computer Internet browser as a result of server response. From this example, we can see that communication between client and server depends on connection speed (bandwidth).

Internet, Intranet and Extranet

- Internet (network of all networks) is a global system comprised of interconnected computers and computer networks, which communicate by means of using TCP/IP protocols. Although, in its beginnings, it emerged from the need for simple data exchange, today it affects all domains of society.
- Intranet is a private network of an organization to which only authorized employees have access (login and password).
- Extranet is part of Intranet, to which independent collaborators have access.

Topic No. 44

Different Way to Connect the Internet

- Dial-up Internet access method uses a modem (56k) and a telephone line.
- Broadband is characterized by a high speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your personal computer.

Different Connection Methods of Internet

Mobile

Internet is connected using a mobile network (GPRS, EDGE, UMTS, HSPA).

Satellite

These are commonly used in parts of the world where there is no proper infrastructure and there is no other way of accessing the Internet.

Wireless (Wi-Fi)

Data is transferred between computers by using radio frequencies (2.4 GHz) and the corresponding antennas.

Cable Internet

It is used in connecting to the Internet through television cable network using a cable modem.

Broadband

It is characterized by a high-speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your personal computer. In the beginning of broadband Internet access, due to underdeveloped communication infrastructure, Internet providers charged based on the data traffic but not time spent on the Internet (unlike dial-up Internet access). Today, in large cities, telecommunications infrastructure is developed, therefore Internet providers do not charge money based on the time spent on the Internet or the amount of transferred data but they do charge by access speed.

Topic No. 45

E-Commerce and E-Banking

- E-commerce is a form of trade that allows customers to browse and purchase products online.
- E-banking allows users to have control over their accounts (view balances and transactions), transactions from one account to another, credit payment, shopping vouchers for mobile phones etc. The benefits are saving time (which would otherwise be spent waiting in lines), lower service fees, and access from anywhere, anytime.

E-Government

- In E-Government, information technology is used to provide better public access to government information, therefore providing citizens with their human right to information.

Different Classification of Projected Aids Projected Aids

- A projected aid is one which items to be observed is projected on a screen using electronic or Mechanical device.
- Very effective because movements can be added to sound and hence a realistic experience can be provided.

Magic Lantern

- It is used for projecting pictures from a transparency on a wall or screen.
- When the figure or illustration is very small and it is required to be shown to the entire class, A transparent slide of the small figure is prepared.
- Then, this slide is placed into the slide carrier part of the magic lantern.
- This magic lantern device projects it on the screen by enlarging its dimension and making the vision more clear and sharp.

Film & Film Projector

- Films are in the form of motion pictures.
- Films enrich learning by presenting a series of meaningful experience involving motion.
- Can enlarge or reduce the actual size of the object to suit the need.
- Film strips & Film strip Projector.
- A film strip is piece of noninflammable safely film, 35 mm wide.
- Length may vary according to requirement and can be up to about one meter.
- One film strip may contain 10 to 50 pictures frames.
- The pictures in film strip may constitute a connected series of drawings, photographs, diagrams or combination of these and illustrated a single sequence.
- The pictures may in color or in black or white.
- Some films strips are accompanied by commentary recorded separately, such film strips is called sound film strip.

Slides & Slide Projector

- The pictures or diagrams which are drawn in suitable medium is called slide. (Glass or Transparency)
- Slide can be enlarged by projecting it on a screen with a slide projector.
- Slide projector is an instrument equipped with a powerful light source and a carrier for holding slides of suitable size.
- Some slide projectors uses drums in which many slides can be loaded in proper sequence in advance.
- It is also possible to record the narration in a tape recorder that could be hooked up to the projector in such a way as to give the necessary commentary without the help of the teacher.

Episcope & Epidiascope

- Episcope is used for the projection of opaque objects.
- Pictures, photographs, drawings or an opaque material within the size of the platform of the episcope can be projected using Episcope.
- Episcope used to enlarge book diagrams and illustrate complex pictures. During Epi-projection the class room should be totally darkened in order to get a clear and bright image.
- Epidiascope can project opaque as well as transparent objects.

OHP

- This is called Over Head Projector because it projects the image behind and over the head of the teacher.
- The transparent visual is placed on a horizontal platform at the top of the light source.
- The teacher can place the transparent plate on the platform and write or draw on it anything that is to be visualized using suitable pen.

Television

- TV is very exciting and efficient means of mass communication.
- Educational Television programme aims at education rather than entertainment.
- One teacher or specialist can render teaching session and pass information to millions of viewers all over the world.

LCD Projector

- LCD (Liquid Cristal Display) is the most advanced and sophisticated projected aid.
- In the area of educational technology, we can replace all other projected and non-projected aids with only a single LCD projector and computer system.
- It can present a topic by using computer made slide, graphs, pictures, video clips, movies and special effect.

DLP Projector (Digital Light processing Projector)

- Most advanced and sophisticated projected aid.
- Brighter image
- Lightest and smallest

Topic No. 46

E-Learning

E-learning consists of all forms of learning and/or knowledge transfer that are based on electronic technologies. This term will mostly be used to describe learning and/or dissemination of knowledge without direct teacher-student contact, while using ICT technology.

Advantages of E-Learning

Advantages of e-learning greatly coincide with learning from books, with a difference that the books come only in text and graphic form on paper, while e-learning materials have multimedia features, and the ability to remotely monitor lectures in real time. To illustrate, physicians can observe medical procedures that are currently taking place in another hospital; students can attend classes that are currently taking place at another university. Some of the advantages for students:

- Adaptive time, place and pace of learning
- Access and availability of education
- Unlimited repetition of lectures
- Reduced cost of education
- Multimedia environment (video, audio, text)

Topic No. 47

Benefits of E-Learning for Teachers and Institutions

Some of the benefits for teachers and educational institutions:

- Reduced costs of conducting teaching (reduced travel costs and accommodation, reduced or eliminated costs for classrooms, etc.)
- Much larger education user base
- Once created, e-education allows educational staff to invest a much shorter time commitment to implement the respective education
- Simple and easy upgrade of materials, etc.

Advantages of Teleworking

- Reduced utility costs and employees transportation reimbursement
- Reduced time for traveling to and from work
- Adapting working conditions to personal habits flexibility
- Possibility of working in the convenience of a person's home (or other places that are not an office)
- Lower stress level, provided there is a good organization and coordination, i.e. the flow of information between employees is unconstrained
- An excellent solution for people with disabilities or long term illness, parents, or those who enjoy traveling

Topic No. 48

E-Mail

E-mail is the transmission of text messages via Internet. Using attachment, users can exchange files like images, multimedia, documents, etc. E-mail addresses consist of a username, @ sign ("at") and a domain name, e.g. mark.marich@ITdesk.info, where "mark.marich" is the username and "ITdesk.info" is the domain name.

Disadvantage of Teleworking

- Physical isolation and lack of direct social contact with colleagues
- Not suitable for everyone - requires a certain dose of self-control and motivation
- Less opportunity for advancement due to lack of traditional vertical hierarchy in the company, however horizontal hierarchy is created, where no one is above anyone, but everyone has their own duties and obligations

Topic No. 49

IM (Instant Messenger)

IM (Instant Messaging) service is used for instant message exchange and real time communication. Besides textual communication and video conferencing, they allow file transfer as well. Some of the popular IM services are: Windows Live Messenger, Skype, and Google Talk.

VoIP (Voice over Internet Protocol)

VoIP (Voice over Internet Protocol) is a term for technology that enables digitalization and transfer of sound (and multimedia sessions) over the Internet. Therefore, it enables sound communication much like the telephone call. In short, VoIP protocol enables audio/video communication between users.

Topic No. 50

RSS (Really Simple Syndication)

RSS (Really Simple Syndication) is a set of web formats, used by web pages that are being constantly updated, considering that by using RSS, new information, articles, etc. can be published. RSS enables us to, with use of appropriate software, read new titles and access news summaries. Users can, with appropriate software, obtain news from various portals and receive information without the need to visit all of them independently, while publishers can distribute news easily to end-users who are monitoring their RSS feeds/channel.

Blog (Web Log)

Blog (web log) is a form of Internet media in which users can re-post or write their opinion on any topic. It can be personal (a form of Internet diary), thematic, or in the form of journals. Blogs are popular because end-users do not need to have advanced computer knowledge, or knowledge of webpage creation, in order to be present on the Internet and write about various topics, i.e. state their opinion. Blogging has greatly facilitated the exchange of interests, ideas, opinions, and contributed to realization of human right to free expression. Some use blogs to share their art, personal aesthetic, or to collect virtual clippings from websites online.

Advantages of Projected Aids

- Teacher can prepare material in advance.
- Saves time in classroom.
- Provides greater enjoyment in learning
- Stimulates more effective learning
- Increases retention
- Increases attention
- Enlarges or reduces actual size of objects.
- Can provide multi-sensory/multimedia approach.
- Brings distant past and the present into the classroom.
- Can make class/presentation more effective.

Topic No. 51

Posts

Entries published on blogs are called posts, which may allow a space below for visitors to leave their comments. Besides text, it is possible to exchange multimedia files, such as photos or videos. In order to share your texts, pictures, video clips, etc., it is necessary to create a personal account on websites that offer this type of service. In case you simply wish to browse the content, it is not necessary to open a personal account.

Podcast

Podcast (POD - Personal on Demand+ Broadcast) is a digital file containing audio or video content. It is distributed over the Internet using RSS technology.

Topic No. 52

Virtual Community

The term "virtual community" encompasses a group of people who communicate via social networks, forums, IM service, blogs. We call it a community, since they are a collection of people who participate in a group of their interests, thus forming a virtual community, because it does not involve physical contact, but is carried out by means of ICT technologies.

Social Networks and Forum

Social Networks

These are free online services that provide users with various forms of presentation and communication. Some of the more popular social networks are: MySpace, Facebook, Twitter, and Google+.

Forum

It is defined as a service that allows users to exchange opinions. It can be compared to bulletin boards on which participants leave their messages, while others leave their comments. Examples of forums are: Reddit, TED Conversations.

Topic No. 53

MMO (Massive Multiplayer Online)

MMO (Massive Multiplayer Online) are network games, played by a large number of players via the Internet. One of the sub-genres of MMO is a MMORPG (Massive Multiplayer Online Role-Playing Game), and its most popular representative is a game called World of War craft.

Chat Room, Video Sharing and Photo Sharing Website

- Chat room is a webpage or part of a webpage which allows a real-time information exchange. Unlike the instant messaging writing software, it does not require computer installation, only a web browser and Internet connection.
- The most popular service for video clip sharing is YouTube.
- Picture or photograph exchange services are Photo bucket, Flickr, Instagram etc.

Topic No. 54

Computer Virus

Virus is a malicious program with the possibility of self-replication. In a computer it seeks other files, trying to infect them and the ultimate goal is to spread to other computers. Depending on what malicious code it contains, it may cause more or less damage to your computer.

Trojan Horses

Trojan horses can be presented in the form of games or files in e-mail attachments, with different purposes, such as: to allow remote control to your computer, delete data, allow spreading of viruses, send spam from the infected computer, etc. Unlike viruses and worms, it does not replicate and it needs to be executed, that is, it is not autonomous.

Worms

Worms like viruses, they possess the ability of self-replication, and however they cannot infect other programs. Their usual intent is to take control of your computer and allow remote control by opening, the so-called, and „backdoor“.

Spyware and Adware

- Spyware is a program that secretly collects data about computer users (passwords, credit card numbers, etc.), therefore it spies on users. The most common spyware infections usually occur while visiting suspicious sites with illegal or pornographic content.
- Adware is a program that uses collected data about a user in order to display advertisements in a way regarded as intrusive. It usually manifests itself in the form of many pop-up windows that display ads.

Disadvantages of Projected Aids

- Expensive.
- Teacher must know to use.(Technical knowledge)
- Requires electricity and other infrastructure.
- Requires a darkroom.
- Requires smart room facilities.
- Chance for diverting from study.
- Require maintenance.
- Require techno friendly (ICT enabled) teaching materials.

Topic No. 55

Hoax

Hoax is a false message being spread by e-mail in order to deceive users. Their goal is to obtain data of, for instance, a bank account, or persuade users to pay some fees in order to win the lottery (which they never applied for), or pay costs of money transfers from an exotic prince to help him withdraw money from a country that is falling apart and, in return, will receive an impressive award, etc.

Process of Software Installation

Installations are available on CD/DVD, USB stick, network drive, the Internet (need to be downloaded), in electronic mail etc. i.e. through any electronic medium capable of storing and distributing digital data.

Topic No. 56

Propriety Software

Propriety software is software licensed under the exclusive legal right of the copyright holder. The licensee is given the right to use the software under certain conditions, but restricted from other uses, such as modification, further distribution etc.

Shareware and Freeware

- Shareware is a term that implies free software use (trials) for a specified time period, after which it is necessary to buy the product.
- Freeware is software free for personal use.

Topic No. 57

WAP (Wireless Application Protocol)

A system that enables you to browse online services, e.g. relating to information about the weather, traffic conditions, shopping, etc. via a special type of mobile phone. WAP is the mobile phone equivalent of the World Wide Web. Newer mobile phones include WAP browser software to allow users access to WAP sites.

Web 2.0

Web 2.0: Contrary to what many people think, Web 2.0 is not a new version of the World Wide Web. The term arose as the name of a series of conferences, the first of which was held in 2004: <http://www.web2summit.com>. Essentially, Web 2.0 is an attempt to redefine what the Web is all about and how it is used, for example new Web-Based communities using Blogs, Podcasts, Wikis and Social Networking websites that promote collaboration and sharing between users - in other words, a more democratic approach to the use of the Web. In order to achieve this, Web based applications have to work more like applications on your computer's hard disc, allowing you to use the Web in much the same way as you would use applications such as Word or PowerPoint. To what extent the concept of Web 2.0 is truly innovative is a matter of debate, as it is broadly in line with the concept of the Web as defined by its inventor.

Topic No. 58

Volatile Memory

It is used to describe the internal main Memory of a computer that loses its contents when power is switched off. RAM is volatile memory as the information is stored in memory chips as an electric charge.

Virtual Reality

The simulation of an environment by presentation of 3D moving images and associated sounds, giving the user the impression of being able to move around with the simulated environment. Users wear helmets and visors that convey the images and sound and gloves that give them the experience of touching objects. The film Lawnmower Man (1992) focused on a character experiencing virtual reality, albeit with negative consequences.

Virtual World

A type of online three-dimensional imaginary world or game in which participants and players adopt amazing characters or avatars and explore the world, engaging in chat or playing complex games. See Avatar, MMORPG, MUVE. See also Section 14.2, Module 1.5 under the heading Chat rooms, MUDs, MOOs and MUVEs. Not to be confused with Virtual Reality, this is a completely different concept.

Topic No. 59

Virtual Learning Environment (VLE)

A VLE is a Web-based package designed to help teachers create online courses, together with facilities for teacher-learner communication and peer-to-peer communication. VLEs can be used to deliver learning materials within an institution or within a local education authority. They may even address a wider constituency, and can even be used on a worldwide basis. VLEs have certain advantages in terms of ease of delivery and management of learning materials. They may, however, be restrictive in that the underlying pedagogy attempts to address a very wide range of subjects, and thus does not necessarily fit in with established practice in language learning and teaching. For this reason some critics argue in favor of a less restrictive Personal Learning Environment (PLE). The two most widely used VLEs in language teaching and learning are Blackboard and Moodle. VLEs may also be referred to as Course Management System (CMS), Learning Management System (LMS), Learning Platform and Learning Support System (LSS).

Video Disc

A technically obsolete Storage Medium, an Optical Disc, 12 inches in diameter, used mainly to store still images or video clips. Now, these are replaced by CD-ROMs and DVDs.

Videodisc Player

Equipment used for accessing information - usually still images or video clips – stored on videodiscs. Now technically obsolete.

Video Memory

The dynamic memory available for the computer's Display Screen. The greater the amount of memory, the greater the possible color depth and resolution of the display.

Topic No. 60**Video Conferencing**

Videoconferencing or Video Conferencing: A computer-based communications system that allows a group of computer users at different locations to conduct a "virtual conference" in which the participants can see and hear one another as if they were in the same room participating in a real conference.

VGA

VGA is abbreviation for Video Graphics Adaptor. An older type of Video Card or circuit board used to control the output on a computer Display Screen. VGA cards were superseded by SVGA cards. See Section 1.1.1.4, Module 1.2, under the heading Graphics card.

Video Card

It is an electronic circuit board inside a computer, which controls the display on the Monitor, i.e. the computer screen. Video cards are usually add-on cards inserted into expansion slots, although sometimes video circuitry is incorporated into the Motherboard. It is usually referred to as a graphics card these days.

Non-Projected Aids**Graphic Aids**

- Graphic Aids are visual aids such as graphs, diagrams, charts etc.
- They can be conveying the message by a combination of written and pictorial visuals that could be made meaningful by suitable captions.
- The criteria for good graphics are that they should be simple, legible and brief.
- Graphs normally used to represent the nature of the relation of two dependent variables. Different types of graphs are Line graph 2. Bar graph 3 Pie graph

Diagrams

- Used to explain many factors at the same time, by using a variety of symbol and labels
- Diagrams can explain facts more vividly than charts

Posters

- Posters are bold and attractive representation of an idea or concept and usually given in color.
- Poster catches the eye and conveys the desired message.

Maps

- A map is an accurate representation of the boundaries and other details of continents, countries etc. on a plane, in the form of a diagram drawn to scale.
- A variety of geographical details like location of mountains and rivers, altitude of places, important cities and other places etc. can be represented accurately with reference to a convenient scale and following suitable color scheme.

Cartoons

- A cartoon is a metaphorical, humorous and exaggerated caricature of a person or situation in the form of a picture or sketch.

Comics

- A comic strip is a form of cartoon depicting a story in sequence.
- The events are arranged in the proper order in a attractive pictorial form that is normally appealing to lower age groups.

Flash cards

- Flash cards are an aid used in language lesson, especially in the lower class.

- The card containing the reading material is exhibited for a few seconds only. Then it is removed and the feedback attempted.

Pictures and Photographs

- Pictures and photographs play a significant role in making ideas clear and comprehensive.